

A COUNTEREXAMPLE TO MARÍN'S CONJECTURE 5.14

ABSTRACT. Conjecture 5.14 of Marín, in arXiv:2608.18302v1, asserts that when the maximal top weight of a certain odd-orthogonal branching expansion is attained by more than one constituent, the tied top contributions cancel. At $(t, r) = (3, 2)$, we show that

$$\beta = (12, 11, 10, 3, 2, 1, 0)$$

has exactly three maximizing constituents with nonzero orthogonal restriction coefficient, namely $(6, 6, 6)$, $(6, 6, 2)$, and $(6, 6, 0)$. Each restriction coefficient is 1 and each top coefficient is ± 1 , so the tied sum is odd and cannot vanish. This refutes the conjecture.

1. THE COUNTEREXAMPLE

We follow the notation of Marín [2]. For a constituent Λ with orthogonal restriction coefficient $a_\Lambda^B \neq 0$, let $T(\Lambda)$ be “the top weight of $c(\Lambda, \cdot)$ delivered by Proposition 5.10(v)” and let

$$c_{\text{top}}(\Lambda) = c(\Lambda, T(\Lambda))$$

be “its value there”. Then

$$M(\beta) = \max\{T(\Lambda) : a_\Lambda^B \neq 0\}$$

is “the largest of them, the maximum taken in the order used by” Marín’s formula for $\mu_{\max}$. Conjecture 5.14 of [2] reads: if $M(\beta)$ is attained by more than one Λ , then

$$(1) \quad \sum_{\Lambda : T(\Lambda) = M(\beta)} a_\Lambda^B c_{\text{top}}(\Lambda) = 0.$$

Four clauses of [2] are used below, and we quote them so that the proof can be followed without the source. Write $t = 2m' + 1$. The entries of $V = 2(\Lambda + \rho_{B_{R'}})$ are sorted into folded residue classes modulo t , of which there are m' nonzero ones and one class 0, and $n_j = \#\{i : V_i \text{ folds to class } j\}$; a set S of indices is a *transversal* when it “contains exactly one index from each nonzero class and none from class 0”. Proposition 5.10 of [2] describes the numerator $\nu(\Lambda, \cdot)$ attached to Λ in these terms: (iii) “ $\nu \equiv 0$ if and only if some $n_j = 0$ — some nonzero residue class is missed by $V = 2(\Lambda + \rho_{B_{R'}})$ ”; (iv) “the value is explicit: at the point indexed by S and a chirality it is $\epsilon_t \det(w) \delta_S \in \{\pm 1\}$ ”, the three factors being those of the source, of which we use only the membership in $\{\pm 1\}$; and (v) “Let $S_{\min}$ be the transversal

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taking, in each nonzero class, the index with the *largest* label — equivalently the smallest value of V . Then the alternant index of $(S_{\min}, +)$ *dominates coordinatewise* the index of every other point of $\text{supp } \nu(\Lambda, \cdot)$, strictly in at least one coordinate. In particular it is the unique maximiser of $\langle w_0, \cdot \rangle$ for every w_0 with all coordinates positive.”

The indexing is the one fixed in the proof of that clause: alternants are indexed “by the doubled vector $2(\mu + \rho_{D_r})$, so that the point of ν attached to $(S, \text{chirality})$ has index $V|_{S^c}$ sorted decreasingly”. At $t = 3$ one has $m' = 1$, so there is a single nonzero folded class, a transversal is a single index, and $S_{\min}$ is the index carrying the smallest entry of V in that class. The top weight attached to Λ is thus read off the vector obtained from $V(\Lambda)$ by deleting that entry, which is the rule used below.

The passage from the numerator to c is a division by the specialised denominator Δ_t , and Proposition 5.19(ii) of [2] inverts it in closed form: with alternants indexed as above and $\tilde{\nu}$, $\tilde{c}$ the $W(D_r)$ -antisymmetric extensions of ν and c , multiplication by Δ_t is “ $a_X \mapsto \sum_{\varepsilon \in \{\pm 1\}^r} (\prod_j \varepsilon_j) a_{X+t\varepsilon}$, and it is inverted by”

$$\tilde{c}(X) = s_r \sum_{k \in (2\mathbb{Z}_{\geq 0} + 1)^r} \tilde{\nu}(X + tk),$$

“a sum with finitely many nonzero terms”, s_r being a global sign. The shifts tk over odd k are the ones compared in the proof below.

Theorem 1. *For $(t, r) = (3, 2)$ and*

$$\beta = (12, 11, 10, 3, 2, 1, 0),$$

one has $M(\beta) = (6, 6)$. The constituents with $a_{\Lambda}^B \neq 0$ and $T(\Lambda) = M(\beta)$ are exactly

$$(6, 6, 6), \quad (6, 6, 2), \quad (6, 6, 0),$$

and each has $a_{\Lambda}^B = 1$. Consequently, the sum in (1) is nonzero.

Proof. Here $N = 7$, $\delta = (6, 5, 4, 3, 2, 1, 0)$, and

$$\lambda = \beta - \delta = (6, 6, 6).$$

Since $\ell(\lambda) = 3 \leq 7/2$ we are in the stable range, where Littlewood’s orthogonal restriction formula [3] gives

$$(2) \quad a_{\Lambda}^B = \sum_{\gamma \in \mathcal{P}^{(2)}} c_{\gamma, \Lambda}^{(6, 6, 6)},$$

where $\mathcal{P}^{(2)}$ is the set of partitions with even parts. Outside the stable range this formula is replaced by a finer branching rule, for which see [1]; the shape at hand does not need it.

For a partition $\Lambda = (\Lambda_1, \Lambda_2, \Lambda_3)$ contained in the rectangle $(6, 6, 6)$, the GL_3 -identity $V_{(6, 6, 6)} = \det^6$ gives the rectangular duality formula

$$c_{\gamma, \Lambda}^{(6, 6, 6)} = \begin{cases} 1, & \gamma = (6 - \Lambda_3, 6 - \Lambda_2, 6 - \Lambda_1), \\ 0, & \text{otherwise.} \end{cases}$$

Since $c_{\gamma, \Lambda}^{(6,6,6)} = 0$ unless Λ is contained in $(6, 6, 6)$ — and Λ has at most $R' = 3$ rows in any case, being a dominant B_3 weight — no Λ outside that rectangle contributes to (2). Thus

$$(3) \quad a_{\Lambda}^B = \begin{cases} 1, & 6 \geq \Lambda_1 \geq \Lambda_2 \geq \Lambda_3 \geq 0 \text{ and all } \Lambda_i \text{ are even,} \\ 0, & \text{otherwise.} \end{cases}$$

For every Λ occurring in (3), set

$$V(\Lambda) = 2(\Lambda + \rho_{B_3}) = (2\Lambda_1 + 5, 2\Lambda_2 + 3, 2\Lambda_3 + 1).$$

At $t = 3$ there is one nonzero folded residue class, represented by $\{\pm 1 \pmod{3}\}$. By Proposition 5.10(iii) of [2], the numerator attached to Λ vanishes only if all three entries of $V(\Lambda)$ are divisible by 3. This cannot occur here. Indeed, if all three entries were divisible by 3, then

$$\Lambda_1 \equiv 2, \quad \Lambda_2 \equiv 0, \quad \Lambda_3 \equiv 1 \pmod{3}.$$

Together with evenness and $0 \leq \Lambda_3 \leq \Lambda_1 \leq 6$, this would force $\Lambda_1 = 2$ and $\Lambda_3 = 4$, a contradiction.

Let $X(\Lambda)$ be the decreasing two-vector obtained by deleting from $V(\Lambda)$ the smallest entry lying in the nonzero folded class. Propositions 5.10(iv),(v) and 5.19 of [2] imply

$$(4) \quad 2(T(\Lambda) + \rho_{D_2}) = X(\Lambda) - 3(1, 1), \quad c_{\text{top}}(\Lambda) \in \{\pm 1\},$$

where $\rho_{D_2} = (1, 0)$.

Indeed, Proposition 5.10(iv),(v) gives the unique coordinatewise maximal numerator index $X(\Lambda)$ with coefficient ± 1 . In the inversion formula of Proposition 5.19(ii), the shift $(1, 1)$ contributes this index at $X(\Lambda) - 3(1, 1)$. Every other shift, after D_2 -straightening, strictly dominates $X(\Lambda)$ coordinatewise and therefore misses the numerator support. The same argument excludes all quotient indices above $X(\Lambda) - 3(1, 1)$.

Since $\Lambda \subseteq (6, 6, 6)$, the entries of $V(\Lambda)$ are strictly decreasing and bounded by $(17, 15, 13)$. Hence

$$X(\Lambda) \leq (17, 15)$$

coordinatewise, and (4) gives

$$T(\Lambda) \leq (6, 6).$$

Equality requires $X(\Lambda) = (17, 15)$, which forces $\Lambda_1 = \Lambda_2 = 6$. Write $\Lambda = (6, 6, k)$. By (3), its restriction coefficient is nonzero exactly for

$$k \in \{0, 2, 4, 6\}.$$

Now

$$V(6, 6, k) = (17, 15, 2k + 1).$$

For $k = 0, 2, 6$, the third entry lies in the nonzero folded class and is smaller than 17, so it is deleted and

$$X = (17, 15).$$

For $k = 4$, the third entry is $9 \equiv 0 \pmod{3}$; the deleted entry is then 17, and

$$X = (15, 9).$$

Therefore the three and only three maximizers with nonzero restriction coefficient are

$$(6, 6, 6), \quad (6, 6, 2), \quad (6, 6, 0).$$

Their restriction coefficients are all 1, while their top coefficients lie in $\{\pm 1\}$ by (4).

The comparisons just made are coordinatewise, whereas the maximum defining $M(\beta)$ is taken in the Weyl majorisation order $\mu \preceq \mu' \iff \mu \in \text{conv}(W(C_r)\mu')$. At $r = 2$ this costs nothing: $\text{conv}(W(C_2)(6, 6)) = [-6, 6]^2$, and the entries of $V(\Lambda)$, hence of $X(\Lambda)$, are positive, so (4) together with $T(\Lambda) \leq (6, 6)$ puts every $T(\Lambda)$ in $[-2, 6]^2$. Thus $T(\Lambda) \preceq (6, 6)$ for every Λ , and $(6, 6)$ is the maximum in that order as well; being the coordinatewise maximum, it also maximises every linear functional with positive coefficients.

The left-hand side of (1) is therefore a sum of three numbers in $\{\pm 1\}$ and cannot be zero. $\square$

Remark 2 (Scope). The hypothesis of Conjecture 5.14 is that $M(\beta)$ be “attained by more than one Λ ”, and $\beta = (12, 11, 10, 3, 2, 1, 0)$ satisfies it. The discussion around the conjecture sometimes phrases the tie case as a top attained twice: Remark 5.13 of [2] writes “Where it is attained twice, the question is whether the tie always cancels”. The dichotomy that remark draws is nevertheless between a top attained once and a top not attained once. Its biconditional, over the 132 shapes tested there, is that the prediction $M(\beta)$ for the top weight of the composite is “correct *exactly* on the 105 where the largest top is attained by a single Λ ”; and the map of conjectures in [2] records that Conjecture 5.14, with Proposition 5.10(v), gives (L2) “*only* where the top is attained once”. A threefold tie is a shape where the top is not attained once, which is where the conjecture is called upon, so the example refutes it as stated.

Remark 3 (The verified population). Conjecture 5.14 is reported in [2] as machine-checked, and $\beta = (12, 11, 10, 3, 2, 1, 0)$ lies outside the population checked. That population is the one of Remark 5.13, “the 132 shapes at $(t, r) = (3, 2), (5, 2), (3, 3)$ ”; the verification table records Conjecture 5.14 there as “27/27 tied shapes; the biconditional 132/132”. Those shapes are the 123 nonvanishing forms of Observation 5.27 — 82 at $(3, 2)$ and 41 at $(5, 2)$ — together with the 9 at $(3, 3)$, and the box of Observation 5.27 is “shapes with $\max \beta \leq 9$ at $(t, r) = (3, 2)$ and $\max \beta \leq 10$ at $(5, 2)$ ”. Here $\max \beta = 12$, so the example contradicts no computation reported in [2].

REFERENCES

- [1] I.-S. Jang and J.-H. Kwon, *Flagged Littlewood–Richardson tableaux and branching rule for classical groups*, J. Combin. Theory Ser. A **181** (2021), Paper No. 105419, doi:10.1016/j.jcta.2021.105419.
- [2] C. Marín, *Schur polynomials twisted by roots of unity and reciprocal pairs: torsion filters, fusion quotients, and an equal-rank reduction at odd order*, arXiv:2608.18302v1 [math.CO] (2026).
- [3] K. Koike and I. Terada, *Young-diagrammatic methods for the restriction of representations of complex classical Lie groups to reductive subgroups of maximal rank*, Adv. Math. **79** (1990), no. 1, 104–135.